

Chapter 22 Electric Fields

Electric Field and Electric Forces

The charge generates an electric field in the space around itself. The basic property of the electric field is to exert a force on other charges in it.

Electric field (electric field strength):

A physical quantity to describe the characteristics of the electric field force.

$$\vec{E} = \frac{\vec{F}}{q_0} \left(\vec{E} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0} \right)$$

It is a vector field. The direction is defined as the electric force of small positive test charge.

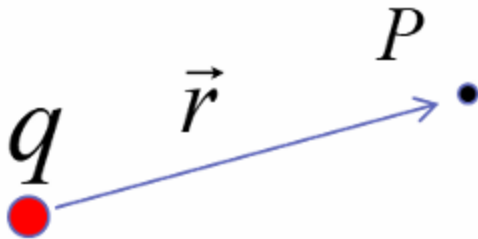
An electric field with equal electric field strength everywhere is called a uniform field.

Electric Field Calculations

Electric Field of Point Charge

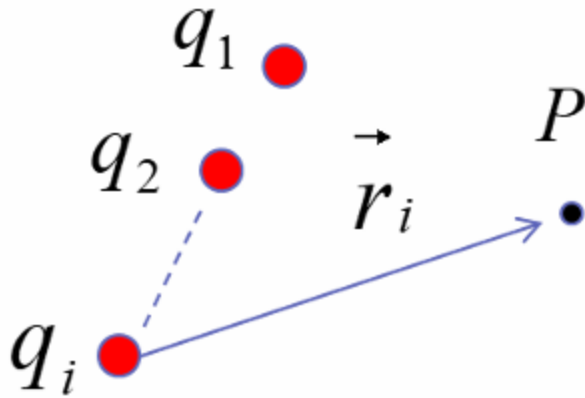
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \hat{r}$$

where $\hat{r} = \frac{\vec{r}}{r}$ is unit vector



Electric Field of Discrete Charges (Principle of Superposition)

$$\vec{E} = \sum \vec{E}_i = \sum \frac{1}{4\pi\epsilon_0} \cdot \frac{q_i}{r_i^2} \hat{r}_i$$



Electric Field of Continuous Charge Distribution

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{r}$$

For electrified body, $dq = \rho dV$; for electrified surface, $dq = \sigma dS$; for electrified line, $dq = \lambda dl$, where ρ, σ, λ is corresponding density of charge.

带电体	带电面	带电线
$\vec{E} = \iiint_V \frac{\rho dV}{4\pi\epsilon_0 r^3} \vec{r}$	$\vec{E} = \iint_S \frac{\sigma dS}{4\pi\epsilon_0 r^3} \vec{r}$	$\vec{E} = \int_L \frac{\lambda dl}{4\pi\epsilon_0 r^3} \vec{r}$
坐标分解, 求 dE_x, dE_y, dE_z		$E_x = \int dE_x$

Problem Solving Strategy

- Apply Coulomb's Law
- Sum all the x- and y- components(convert the vector to scalars)
- Find the resultant force

Problem-Solving Tactics-- A Field Guide for Lines of Charge

General strategy is to pick out an element dq of the charge, find $d\vec{E}$ due to the element and integrate $d\vec{E}$ over the entire line of charge.

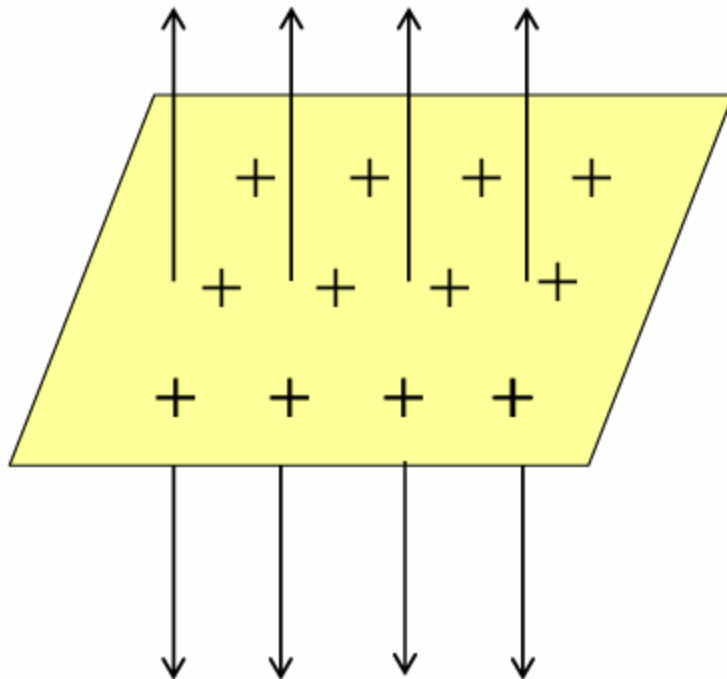
Note for the symmetry situation.

- Rings(P on axis of symmetry)
Replace r^2 with $z^2 + R^2$. Express the adding component of $d\vec{E}$ in terms of θ where θ is the angle between the electric field and the symmetric axis. And $\cos \theta = \frac{z}{r}$.
- Circular arc(P at the center of curvature)
Express the adding component of $d\vec{E}$ by θ and use $ds = r d\theta$ transfer the variable to θ .
- Straight line(P on an extension of the line)
Replace r with x .
- Straight line(P at perpendicular distance from the line of charge)
Replace r with x and y . If P is on the perpendicular bisector, replace the variable by trigonometric function. If not, respectively integrate x- and y- components and then integrate the total electric field.

Expand

For a flat plate with infinite charges evenly distributed in space, it generates a uniform field and

$$E = \frac{\sigma}{2\epsilon_0}$$



For parallel capacitor, it can be seen as two flat plate with infinite charges. The electric field outside is zero, inside is

$$E = \frac{\sigma}{\epsilon_0}$$

Electric Field Lines

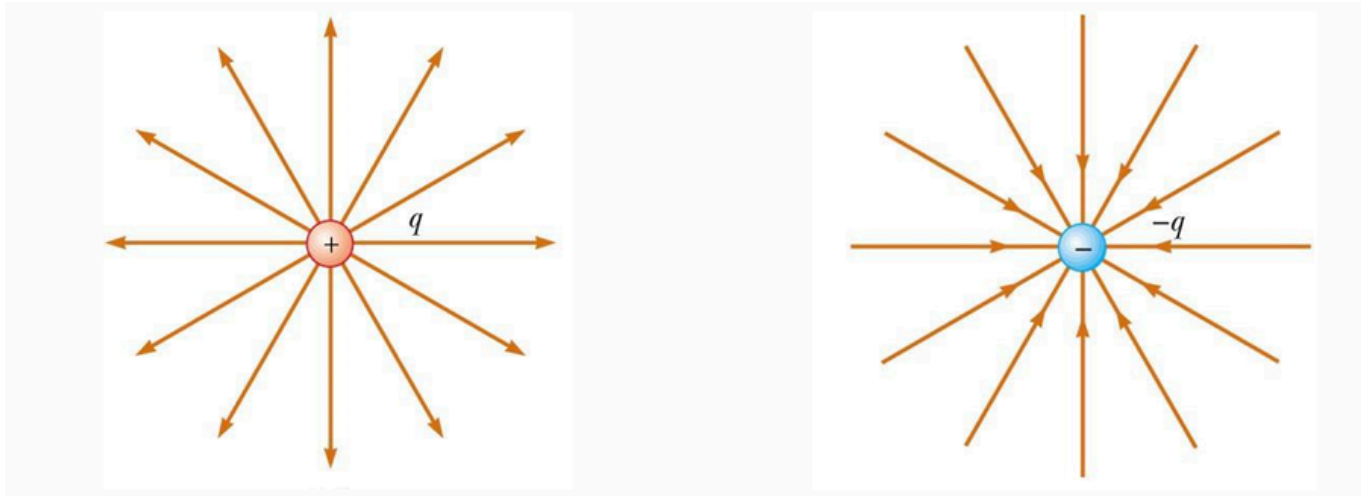
The tools to describe the spatial distribution of the electric field.

The direction of the tangent line represents the direction of the electric field strength at that point, and the density of the curves represents the magnitude of the electric field strength at that point.

Characteristic of electric field lines:

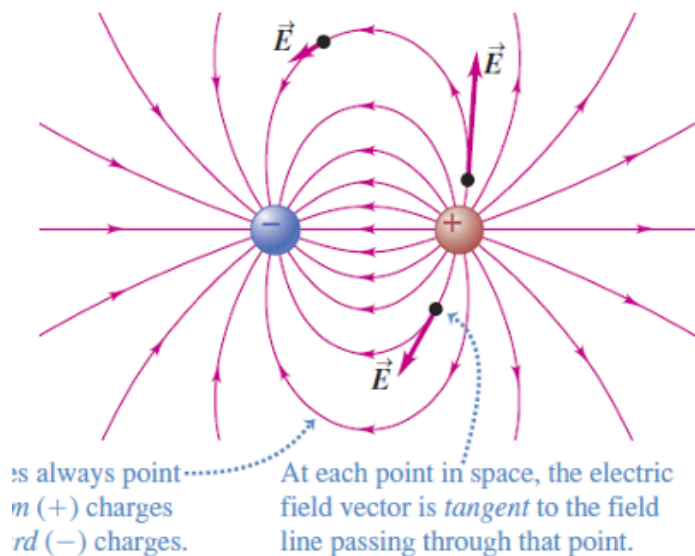
- Starting from a positive charge and returning to a negative charge, the curve is an open line.
- Electric field lines are artificially introduced and do not actually exist.

For point charge, the lines radiate equally in all directions(positive outward, negative inward)



For two equal charges

(b) Two equal and opposite charges (a dipole)



(c) Two equal positive charges

